

Technology Stack

Date	29 June 2026
Team ID	
Project Name	Smart Lender project
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap 5, JavaScript, Jinja2 Templates	Provides a responsive, interactive, and user-friendly interface for loan prediction, dashboard visualization, and report generation.

2	Backend / Server-Side	Python, Flask Framework, Scikit-learn	Flask handles routing, authentication, and request processing, while Scikit-learn integrates the trained Random Forest model for loan eligibility prediction.
3	Database / Data Storage	SQLite	Lightweight relational database used to securely store user accounts, prediction history, applicant details, and generated reports.
4	Cloud / Hosting / Deployment	Render	Provides simple cloud deployment, automatic GitHub integration, HTTPS support, and reliable hosting for the Flask application.
5	Version Control & CI/CD	Git, GitHub	Enables source code management, version tracking, collaboration, backup, and automatic deployment through Render.
6	Third-Party APIs / Other Tools	ReportLab, Pickle, NumPy, Pandas, Random Forest Model	ReportLab generates downloadable PDF reports, Pickle loads the trained machine learning model, NumPy and Pandas process applicant data, and the Random Forest model performs loan eligibility prediction.